

LINKEDLIST

Q1)

Question:

DS	Session	Linked List	Question Information	Level 1 Challenge 31
Question description Professor Shiva decided to conduct an industrial visit for final year students, but he set a condition that if students received a passing grade in the surprise test, they would be eligible to go on the industrial visit. He asked the students to study a topic linked list for 10 minutes before deciding to conduct a surprise test. Professor-mandated questions, such as the deletion of nodes with a certain data D, are now being asked. For example if the given Linked List is 5->10->15->10->25 and delete after 10 then the Linked List becomes 5->15->25. Constraints 1 < N < 100 1 < D < 1000 Input Format First line contains the number of datas- N. Second line contains N integers the given linked list . Next line indicates the node data D that has to be deleted. Output Format Single line represents the linked list after required elements deleted.				

PROGRAM:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct Node {
5      int data;
6      struct Node *next;
7  };
8
9  struct Node* createNode(int data) {
10     struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
11     newNode->data = data;
12     newNode->next = NULL;
13     return newNode;
14 }
15
16 struct Node* insertEnd(struct Node *head, int data) {
17     struct Node *newNode = createNode(data);
18
19     if (head == NULL)
20         return newNode;
21
22     struct Node *temp = head;
23
24     while (temp->next != NULL)
25         temp = temp->next;
26
27     temp->next = newNode;
28
29     return head;
30 }
31
32 struct Node* deleteData(struct Node *head, int D) {
33     while (head != NULL && head->data == D) {
34         struct Node *temp = head;
35         head = head->next;
36         free(temp);
37     }
38
39     if (head == NULL)
40         return head;
41
42     struct Node *temp = head;
43
44     while (temp->next != NULL) {
45         if (temp->next->data == D) {
46             struct Node *deleteNode = temp->next;
47             temp->next = temp->next->next;
48             free(deleteNode);
49         } else {
50             temp = temp->next;
51         }
52     }
53
54     return head;
55 }
```

```
7  void display(struct Node *head) {
8      printf("Linked List:");
9
10     while (head != NULL) {
11         printf("->%d", head->data);
12         head = head->next;
13     }
14
15     printf("\n");
16 }
17
18 int main() {
19     int N, D, data;
20     struct Node *head = NULL;
21
22     scanf("%d", &N);
23
24     for (int i = 0; i < N; i++) {
25         scanf("%d", &data);
26         head = insertEnd(head, data);
27     }
28
29     scanf("%d", &D);
30
31     head = deleteData(head, D);
32
33     display(head);
34
35     return 0;
36 }
```

OUTPUT:

TEST CASE 1:

```
6
7 9 8 8 4 2
8
Linked List:->7->9->4->2
```

TEST CASE 2:

```
10
5 3 7 2 7 9 8 8 4 2
2
Linked List:->5->3->7->7->9->8->8->4
```